UNIVERSITY OF CAPE TOWN

Test - May 2023

Investment Decision Theory and Application
ECO4102Z/ECO4122Z

Time: 2 hours 30 minutes

Maximum Marks: 80

INSTRUCTIONS TO CANDIDATES

1. You have 10 minutes at the start of the examination in which to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made, on this examination paper. You then have three hours 30 minutes to complete the paper.
2. You must not start writing your answers in the booklet until instructed to do so by the invigilator.
3. Mark allocations are shown in brackets [ ].
4. Attempt all 8 questions, beginning your answer to each full question on a new page.
5. Show all calculations where this is appropriate.
6. Write legibly using blue or black ink.

1. Nigel Mpinyuri, aged 40, is a store manager at Pic n' Pay and earns an annual salary of R1,440,000 before taxes. Eveline Mpinyuri, aged 38, stays home to care of their newborn twins. She recently inherited R16,200,000 (after wealth-transfer taxes) in cash from her father's estate. In addition, the Mpinyuris have accumulated the following assets (current market value):

   - R90,000 in cash.
   - R2,880,000 in stocks and bonds.
   - R3,960,000 in Pic n' Pay equity shares.

   The value of their holdings in Pic n' Pay shares has appreciated substantially as a result of the company's growth in sales and profits during the past 10 years. Nigel Mpinyuri, is confident that the company and its stock will continue to perform well. The Mpinyuris need R540,000 for a down payment on the purchase of a house and plan to make a R360,000 non-tax-deductible donation to a local charity in memory of Eveline Mpinyuri's father. The Mpinyuris' annual living expenses are R1,332,000. After-tax salary increases will offset any future increases in their living expenses. During their discussions with their financial advisor, Greg Bolus, the Mpinyuris express concern about achieving the educational goals of their children and their own retirement goals. The Mpinyuris tell Greg:

   - They want to have sufficient funds to retire in 18 years when their children begin their four years of university education.
   - They have been unhappy with the portfolio volatility they have experienced in recent years and they do not want to experience a loss greater than 12% in any one year.
   - They do not want to invest in alcohol and tobacco stocks.
   - They will not have any additional children.

   After their discussions, Greg calculates that in 18 years the Mpinyuris will need R36 million to meet their educational and retirement goals. Greg suggests that their portfolio be structured to limit shortfall risk (defined as expected total return minus two standard deviations) to no lower than a -12% return in any one year. Mpinyuri's salary and all capital gains and investment income are taxed at 40% and no tax-sheltering strategies are available. Greg's next step is to formulate an investment policy statement for the Mpinyuris.

   (a) Formulate the risk objective of an investment policy statement for the Mpinyuris. [5 marks]

   (b) Formulate the return objective of an investment policy statement for the Mpinyuris Calculate the pretax rate of return that is required to achieve this objective. Show your calculations. [9 marks]

   (c) Formulate the constraints portion of an investment policy statement for the Mpinyuris, addressing each of the following:

Question 1 continues on the next page...

i. Time horizon

ii. Liquidity requirements

iii Unique circumstances.

[6 marks]

[Total: 20 marks]

SUGGESTED SOLUTION

a. The Mpinyuris' overall risk objective must consider both willingness and ability to take risk:
Willingness: The Mpinyuris, have a below-average willingness to take risk, based on their unhappiness with the portfolio volatility they have experienced in recent years and their desire not to experience a loss in portfolio value in excess of 12 percent in any one year.-[maximum 2 marks]
Ability: The Mpinyuris have an average ability to take risk. Although their fairly large asset base and long time horizon in isolation would suggest an above-average ability to take risk, their living expenses of R1,332,000 are significantly higher than Nigel's after-tax salary of R1,440,000(1-0.40)= 864,000 causing them to be very dependent on projected portfolio returns to cover the difference and thereby reducing their ability to take risk. [maximum 2 marks]
Overall: The Mpinyuris' overall risk tolerance is below average, as their below-average willingness to take risk dominates their average ability to take risk in determining their overall risk tolerance. [maximum 1 mark]

b. The Mpinyuris' return objective is to grow the portfolio to meet their educational and retirement needs as well as to provide for ongoing net expenses. The Mpinyuris will require annual after-tax cash flows of R468,000 (calculated below) to cover ongoing net expenses and will need R36 million in 18 years to fund their children's education and their retirement. To meet this objective, the Mpinyuris' pretax required return is 7.38 percent which is determined below. The after-tax return required to accumulate R36 million in 18 years beginning with an investable asset base of R22,230,000 (calculated below) and with annual outflows of R468,000 is 4.427 percent, which when adjusted for the 40 percent tax rate, results in a 7.38 percent pretax return (4.427%/(1 - 0.40)= 7.38%).

Nigel's Annual Salary	1,440,000
Less: Taxes (40%)	(576,000)
Living Expenses	(1,332,000)
Net Annual Cash Flow	**(468,000)**
Inheritance	16,200,000
Pic n' Pay shares	3,960,000
Stocks and Bonds	2,880,000
Cash	90,000
Subtotal	**23,130,000**
Less One-time Needs:	
Down Payment on House	(540,000)
Charitable Donation	(360,000)
Investable Asset Base	**22,230,000**

Note: No inflation adjustment is required in the return calculation because increases in living expenses will be offset by increases in Nigel's salary.

c **Time horizon.** The Mpinyuris have a two-stage time horizon, because of their changing cash flow and resource needs. The first stage is the next 18 years. The second stage begins with their retirement and the university education years for their children. [maximum 2 marks]
ii. **Liquidity requirements.**The Mpinyuris have one-time immediate expenses totalling R900,000 that include the deposit on the house they are purchasing and the charitable donation in honour of Eveline's father. [maximum 2 marks]
iii **Unique circumstances.** The large holding of the Pic n' Pay equity shares represents almost 18 percent of the Mpinyuris' investable asset base. The concentrated holding in Pic n' Pay stock is a key risk factor of the Mpinyuris' portfolio and achieving better diversification will be a factor in the future management of the Mpinyuris' assets. The Mpinyuris' desire not to invest in alcohol and tobacco stocks is another constraint on investment. [maximum 2 marks]

2. (a) Explain clearly the difference between a float-adjusted market-capitalization-weighted index and a market-value-weighted index. [2 marks]

   (b) State any four(4) categories of shares that are included in the free float restrictions when calculating the FTSE/JSE Africa Index Series. [2 marks]

   Consider the three stocks in the following table. P_t represents price, in Rand, at time t, and Q_t represents shares outstanding at time t. Stock C splits two for one in the last period.

	P_0	Q_0	P_1	Q_1	P_2	Q_2
A	90	100	95	100	95	100
B	50	200	45	200	45	200
C	100	200	110	200	55	400

(c) Calculate the rate of return on a price-weighted index of the three stocks for the first period ($t = 0$ *to* $t = 1$). [2 marks]

(d) Determine the divisor for the price-weighted index in period 2. [1 marks]

(e) Calculate the rate of return for the second period ($t = 1$ *to* $t = 2$). [2 marks]

[**Total: 9 marks**]

At $t = 0$, the value of the index is: $(90 + 50 + 100)/3 = 80$
At $t = 1$, the value of the index is: $(95 + 45 + 110)/3 = 83.333$.
The rate of return is: (83.333/80)-1=4.17%
b. In the absence of a split, Stock C would sell for 110, so the value of the index would be: $250/3 = 83.333$ After the split, Stock C sells for 55. Therefore, we need to find the divisor (d) such that: 83.333 = (95 + 45 + 55)/d , d = 2.340
c. The return is zero. The index remains unchanged because the return for each stock separately equals zero.
Some students calculated a value-weighted index. some did not weight the index

3. A risk averse investor makes decisions using a quadratic utility function:

$$U(w) = w + dw^2.$$

(a) Derive an upper bound for d for this investor. [2 marks]

(b) Explain why the investor can only use this utility function to make decisions over a limited range of wealth, w. Your answer should include a statement of this range.[3 marks]

The investor states that the upper limit of wealth where she can use this utility function is w = R1,000.

(c) Determine the value of d in the investor's utility function. [1 mark]

The investor wins a prize of R250 in a gameshow. She is then offered the opportunity to exchange this prize for a larger prize of R600 if she can answer one more question correctly. However, she will receive no prize at all if she gets the question wrong. She estimates her chances of answering the question correctly to be 50%.

(d) Determine whether the investor should take this opportunity to exchange. [3 marks]

[**Total: 9 marks**]

(a)

$$U'(w) = 1 + 2dw, [1/2]$$

and

$$U''w) = 2d.[1/2]$$

Because the investor is risk averse, we must have $U''(w) < 0$ (alternatively to satisfy the condition of diminishing marginal utility of wealth (risk aversion)) [1/2]
So we must have $d < 0$. [1/2]
(b) The condition of non-satiation requires $U'(w) > 0$. [1]
Hence $1 + 2dw > 0$ and $w < -1/(2d)$ [1]
So the quadratic utility function can only satisfy the condition of non-satiation over a limited range of w:
Specifically $-\infty < w < -1/(2d)$ [1]
(c)

$$\begin{aligned} 1,000 &= -1/2d \quad [1/2] \\ \Rightarrow d &= -1/2000 = -0.0005 \quad [1/2] \end{aligned}$$

(d)

$$\begin{aligned} U(250) &= 250–0.0005 \times 2502 = 218.75 \quad [1] \\ E[U(\text{exchange})] &= 0.5 \times U(600) + 0.5 \times U(0)[1/2] \\ &= 0.5 \times (600–0.0005 \times 6002) = 210 \quad [1/2] \end{aligned}$$

So the investor should not accept the opportunity to exchange… [$\frac{1}{2}$]
… because the expected utility of the exchange opportunity is lower than that of the prize. [$\frac{1}{2}$]
A number of students struggled with this. However those who got it did very well, in all sections

4. (a) Define any one of the following terms:
   i. Mean-preserving spread
   ii. Second-order stochastic dominance

[1 mark]

Consider the following investments whose expected returns(%), have the following probability distribution(s):

	Probability	
Return	**Investment 1**	**Investment 2**
10%	0.25	0.20
12%	0.25	0.50
14%	0.25	0.10
15%	0.25	0.20

(b) Compare the two investments using the second-order stochastic dominance criterion.

[5 marks]

[Total: 6 marks]

a. a gamble X is said to be a mean preserving spread of Y if the two have the same expectation and Y second-order stochastically dominates X- some stduents may use a graph to illustrate
b. No dominance,. students to show calculations
Some students made the correct calculations but came to a wrong conclusion

5. Suppose a market with three risky assets: X, Y, and Z with the following properties.

	Expected Return(%)	Standard Deviation(%)
Asset X	10	4
Asset Y	12	10
Asset Z	18	14

The risk-free rate is 5 percent, and the correlation between assets X and $Y(\rho_{xy})$ is 0.50, the correlation between assets X and $Z(\rho_{xz})$ is 0.71 and the correlation between assets Y and $Z(\rho_{yz})$ is 0.50.

(a) Determine the expected return and standard deviation of a portfolio of X, Y and Z that maximizes the Sharpe ratio (i.e. optimal portfolio). Assume no restrictions to short selling. [10 marks]

(b) Investors can hold any portfolio consisting of asset X, Y and Z and also the risk-free asset. In addition, suppose the Satrix Top40 fund offers an expected return of 9% and a standard deviation of 6%. All investors have standard mean-variance preferences. Should any investor invest in the Satrix Top40 fund? Explain clearly. [3 marks]

[Total: 13 marks]

w1=0,959291822 ; w2=0,029868722; w3=0,010839456; portfolio mean = 0,10146 portfolio sigma =0,04106 $\sigma_p = 0.24$-*1 mark*
Should invest not invest- has lower return but higher variance than the optimal portfolio, therefore it is inefficient.
A few students made a mistake on the covariance and used the correlations instead of variance

6. In a market where the CAPM holds there are five risky assets with the following attributes per year.

Asset number	1	2	3	4	5
Expected return	11%	10%	13%	18%	16%
Market capitalisation (in ZAR)	2.6bn	3.9bn	5.2bn		1.3bn
Beta				1.5	

The risk-free rate is $r_f = 6\%$ p.a.

(a) Calculate the expected return on the market portfolio. [2 marks]

Question 6 continues on the next page. . .

(b) Deduce the market capitalisation of asset 4 and the betas of all the other assets. [6 marks]

(c) Calculate the beta of a portfolio P which is equally weighted in the five assets and the risk-free asset. [2 marks]

[Total: 10 marks]

From CAPM, we have

$$(a).R_4 = 18\% = 6\% + 1.5(R_m - 6\%)[1] \Rightarrow R_m = 14\%[1]$$

(b).

Asset number	1	2	3	4	5
Expected return	11%	10%	13%	18%	16%
Market capitalisation (in ZAR)	2.6bn	3.9bn	5.2bn	6.5bn	1.3bn
Beta	0.625	0.5	0.875	1.5	1.25

Note: Market Cap obtained as follows: $2.6(11) + 3.9(10) + 5.2(13) + 18x + 13(16) = 14(13(13 + x) \Rightarrow x = 6.5$- [1]
Beta obtained as follows; Since the CAPM holds we $\beta = \frac{R_i - R_f}{R_m - R_f}$
(c),. $\beta_p = \sum_{i=1}^{6} w_i \beta_i = 0.79167$ or 19/24
Some students forgot that the portfolio in (c) will have 6 assets, therefore the weight of each is 1/6; they used 1/5

7. Liam Truter is a portfolio manager for a bank trust department. Truter meets with two clients, Travis Bradfield and Ruth Nonde, to review their investment objectives. Each client expresses an interest in changing his or her individual investment objectives. Both clients currently hold well-diversified portfolios of risky assets.

   (a) Travis wants to increase the expected return of his portfolio. State what action Truter should take to achieve Travis's objective. Justify your response in the context of the CML. [2 marks]

   (b) Ruth wants to reduce the risk exposure of her portfolio but does not want to engage in borrowing or lending activities to do so. State what action Truter should take to achieve Ruth's objective. Justify your response in the context of the SML. [3 marks]

[Total: 5 marks]

a. Truter should borrow funds and invest those funds proportionately in Travis' existing portfolio (i.e., buy more risky assets on margin). In addition to increased expected return, the alternative portfolio on the capital market line will also have increased risk, which is caused by the higher proportion of risky assets in the total portfolio.
b. Truter should substitute low beta stocks for high beta stocks in order to reduce the overall beta of Ruth's portfolio. By reducing the overall portfolio beta, Truter will reduce the systematic risk of the portfolio, and therefore reduce its volatility relative to the market. The security market line (SML) suggests such action (i.e., moving down the SML), even though reducing beta may result in a slight loss of portfolio efficiency unless

full diversification is maintained. Ruth's primary objective, however, is not to maintain efficiency, but to reduce risk exposure; reducing portfolio beta meets that objective. Because Ruth does not want to engage in borrowing or lending, Truter cannot reduce risk by selling equities and using the proceeds to buy risk-free assets (i.e., lending part of the portfolio)

A number of students ignored the advise to use the SML and the CML, leading to incorrect decisions.

8. Denisha Ramaloo, a portfolio manager at Mashilo Asset Management, is using the capital asset pricing model for making recommendations to her clients. Her research department has developed the information shown in the following exhibit.

	Forecast Return	Standard Deviation	Beta
Stock X	14.0%	36%	0.8
Stock Y	17.0%	25%	1.5
Market index	14.0%	15%	1.0
Risk-free rate	5.0%		

(a) Calculate expected return and alpha for each stock. [4 marks]

(b) Identify and justify which stock would be more appropriate for an investor who wants to

i. Add this stock to a well-diversified equity portfolio.

ii. Hold this stock as a single-stock portfolio.

[4 marks]

[**Total: 8 marks**]

a. Expected return and alpha for each stock

	Expected Return	Alpha
Stock X	5% + 0.8(14%-5%)=12.2%	14.0%-12.2% = 1.8%
Stock Y	5% + 1.5(14% -5%)=18.5%	17.0%-18.5% =-1.5%

b. i. Denisha should recommend Stock X because of its positive alpha, compared to Stock Y, which has a negative alpha. In graphical terms, the expected return/risk profile for Stock X plots above the security market line (SML), while the profile for Stock Y plots below the SML. Also, depending on the individual risk preferences of Kay's clients, the lower beta for Stock X may have a beneficial effect on overall portfolio risk.

b ii. Denisha should recommend Stock Y because it has higher forecasted return and lower standard deviation than Stock X. Some students may proceed to calcute the respective Sharpe ratios for Stocks X and Y and the market index are:
Stock X: (14%-5%)/36%= 0.25
Stock Y: (17%-5%)/25% = 0.48
Market index: (14%- 5%)/15%= 0.60

 Question 8 continues on the next page...

The market index has an even more attractive Sharpe ratio than either of the individual stocks, but, given the choice between Stock X and Stock Y, Stock Y is the superior alternative. When a stock is held as a single stock portfolio, standard deviation is the relevant risk measure. For such a portfolio, beta as a risk measure is irrelevant. Although holding a single asset is not a typically recommended investment strategy, some investors may hold what is essentially a single-asset portfolio when they hold the stock of their employer company. For such investors, the relevance of standard deviation versus beta is an important issue.

Most student managed to get the first part(a). A number of struggled with b(ii)
